Reflexive and complementary dual codes from graph families

1 Source and common interfaces

This application concerns the complete mathematical result set of J. D. Key and B. G. Rodrigues, *LCD codes from adjacency matrices of graphs*, author manuscript dated 4 August 2017 (F017, seventeen pages). The source's constructions, classifications, finite tables and recorded questions remain distinguished. Several statements as printed need correction; the qualifications below concern this exact edition and do not attribute its errors to earlier papers without examining them.

Throughout, a graph is finite, simple and undirected, with at least one vertex. Over $\mathbb{F}_p$, let A be its adjacency matrix and set

$$B = J-I-A, \qquad R = I+A, \qquad \bar{R} = I+B = J-A, \quad C = \operatorname{im} A, \quad \bar{C} = \operatorname{im} B, \quad RC = \operatorname{im} R.$$

An orthogonal complement is for the usual coordinate form. A code is LCD if its intersection with its orthogonal complement is zero. Following the source, C is RLCD if $C = (RC)^{\perp}$; for a regular graph it is CLCD if $C = \bar{C}^{\perp}$. These stronger properties should not be confused with LCD itself.

The complete common paper supplies the orthogonal-projector theorem, its exact constant-line defect, the scalar intersection-count criterion, the simplex boundary argument, the ternary inclusion argument and the transitive systematic-message certificate. The exact dependency map binds their previously reviewed proof texts and required computations. They are actual prerequisites, not free references to an uncharged proof bank. This application supplies the source-specific substitutions, complete output tables and qualifications. The common paper and all required finite certificates must accompany it in a complete edition. The classical spectra and some distances explicitly quoted in the source from earlier work remain separately attributed inputs.

2 Projection and decoding statements

For any linear code $C \leq \mathbb{F}_p^N$, the condition $C + C^{\perp} = \mathbb{F}_p^N$ is equivalent to $C \cap C^{\perp} = 0$, by the dimension formula. This is source Lemma 1. The shared symmetric-idempotent argument gives the actual orthogonal projections, rather than just a rank test.

If $A(A + I) = 0$, the projections onto C and RC are $-A$ and $I + A$. Conversely $C \perp RC$ implies this identity, so containment already implies $C = (RC)^{\perp}$. This recovers source Proposition 1(1), Proposition 2(1) and quoted Result 2 from Proposition 1.4 of the common paper. The unconditional inclusion in Result 2 is also immediate: if $Av = 0$, then $v = (A+I)v$, so $C^{\perp} \subseteq RC$. For a received row vector $v = \sum_x v(x)e_x$, with adjacency and reflexive rows r_x, s_x, respectively, the decomposition is

$$v = -\sum_x v(x)r_x + \sum_x v(x)s_x.$$

In characteristic two, if v already belongs to one summand, it is the sum of its corresponding rows indexed by $\operatorname{Supp}(v)$.

Suppose the graph is regular of degree ν and $AB = 0$. Put $\bar{\nu} = N - 1 - \nu$, with these integers also denoting their field images. Theorem 1.2 of the common paper states that both codes are LCD and gives

	orthogonal decomposition	projection matrices
$\nu \neq 0$	$\mathbb{F}_p^N = C \perp\!\!\oplus \bar{C}$	$J - A,\ -B$
$\bar{\nu} \neq 0$	$\mathbb{F}_p^N = C \perp\!\!\oplus \bar{C}$	$-A,\ J - B$
$\nu = \bar{\nu} = 0$	$\mathbb{F}_p^N = C \perp\!\!\oplus \bar{C} \perp\!\!\oplus \langle \mathbf{j} \rangle$	$-A,\ -B,\ J.$

Thus the exceptional case in source Proposition 2(2) occurs exactly when both degrees vanish. Its defect is exactly one and is the displayed constant line; no integral-spectrum assumption is necessary. When the codes are complementary this is precisely Proposition 1(2), including its distinction according to whether $p \mid \nu$.

The relation between the two reflexive tests is

$$B(B + I) = A(A + I) + (N - 2\nu - 1)J. \tag{1}$$

Consequently, given that either graph is RLCD, the other is RLCD exactly when $N-2\nu-1 = 0$ in the field. The congruence alone does not establish RLCD: for the five-cycle over $\mathbb{F}_2$ it holds, but adjacent vertices have no common neighbour, so the corresponding entry of $A(A + I)$ is one. Source Lemma 2 must be read as a compatibility statement or corrected by adding the premise that one graph is RLCD.

For integral spectra, Lemma 8.3 of the common paper proves source Lemma 3: once C is LCD, its dimension is the sum of the multiplicities of eigenvalues nonzero modulo p. The proof uses the characteristic polynomial of the actual integer matrix together with its LCD image/kernel decomposition; it does not assume that reduction of a real eigenbasis is a basis over $\mathbb{F}_p$. The preliminary spectral observation on source page 5 also has its precise elementary justification. For a symmetric integer matrix with distinct integral eigenvalues λ_i, the integer polynomial $\prod_i (X - \lambda_i)$ annihilates the matrix, since it does over $\mathbb{R}$. If none reduce to zero, reduction gives invertibility. If exactly one distinct λ_i reduces to zero, the reduced annihilator has only a simple factor X, so the generalized zero eigenspace is the kernel. Its dimension is the multiplicity of zero in the characteristic polynomial, namely the multiplicity of that λ_i. With several eigenvalues reducing to zero, only the general lower rank bound obtained from algebraic multiplicity is automatic; the LCD premise supplies the exact formula used here.

Finally the decoding assertion quoted as source Result 1 is Massey's nearest-neighbour observation. If $v = v_C + v_\perp$ is the orthogonal splitting and $u \in C$ minimizes $\mathrm{wt}(v_\perp - u)$, then $v_C + u$ is a nearest word in C, since translation by v_C identifies the two minimization problems. This is a statement about Hamming distance and does not require an orthogonal sum formula for that distance. It gives the source's decoder once the projections above are known.

3 Uniform subset graphs

Let Ω have n elements and let $\Gamma(n, k, r)$ join two k-subsets when their intersection has size r, where $0 \leq r < k$. Unless an endpoint is explicitly discussed, take $n \geq 2k + 1$ and $k \geq 2$. Write $N = \binom{n}{k}$, $\nu = \binom{k}{r}\binom{n-k}{k-r}$ and $\bar{\nu} = N - 1 - \nu$. A binomial coefficient outside its ordinary range is zero. For $|x \cap y| = j$, put

$$i_j = \sum_a \binom{j}{a}\binom{k-j}{r-a}^2\binom{n-2k+j}{k-2r+a}, \qquad P_j = i_j + [j = r], \qquad \ell_j = \nu - P_j.$$

The natural S_n action is transitive on vertices and on each ordered pair class of fixed intersection size. Thus it is transitive on the nonloop edges of $\Gamma(n, k, r)$. The complement

generally contains several such classes, and loops form a separate class in the reflexive graph; the source's phrase extending edge transitivity to both associates cannot be taken literally for this S_n action. No such additional transitivity is used in the count argument. These are exactly the common-neighbour and scalar products in source Proposition 3. The four-region count is proved in Equation (5.1) of the common paper; Theorem 5.1 of the common paper gives

$$\begin{aligned} C \text{ is RLCD} &\iff P_j = 0 \text{ for all } j, \\ C \perp \bar{C} &\iff P_j = \nu \text{ for all } j, \\ C \text{ is CLCD} &\iff P_j = \nu \text{ for all } j \text{ and } (\nu, \bar{\nu}) \neq (0,0), \\ \bar{C} \text{ is RLCD} &\iff \bar{\nu} - \nu + P_j = 0 \text{ for all } j. \end{aligned} \tag{2}$$

All equalities in these tests are in $\mathbb{F}_p$. In the last line the diagonal test forces $\bar{\nu} = 0$, so $C \perp \bar{C}$. If also $\nu = 0$, then C is RLCD; otherwise $C = \bar{C}^{\perp} = \operatorname{im}(I + B)$. This supplies all three clauses of the source proposition, with its possible complement exception replaced by the exact constant-line defect.

For the Johnson graph $r = k-1$, the only possibly nonzero i_j are $i_{k-2} = 4$, $i_{k-1}+1 = n-1$ and $i_k = k(n-k)$. For the Kneser graph $r = 0$, $i_j = \binom{n-2k+j}{k}$. For $r = 1$ they are

$$i_j = j\binom{n-2k+j}{k-1} + (k-j)^2\binom{n-2k+j}{k-2}.$$

Together with $\ell_j = \nu - P_j$, these give every formula in source Example 2. Only realized intersection sizes should be tested outside the domain $n \geq 2k+1$.

Corollary 1. *In this domain, the complete conditions for the five families listed on source page 11 are the following. A blank condition means that there are no further primes or residues.*

(k, r)	*RLCD*	*CLCD*
$(2,0)$	$p = 2,\ n \equiv 3 \pmod 4$	$p = 2,\ n \equiv 1 \pmod 4$
$(3,0)$	$p = 2,\ n \equiv 1 \pmod 4$	$p = 2,\ n \equiv 1 \pmod 4$
	$p = 3,\ n \equiv 5 \pmod 9$	$p = 3,\ n \equiv 2, 8 \pmod 9$
	$p \geq 5,\ n \equiv 5 \pmod p$	$p \geq 5,\ n \equiv 5 \pmod p$
$(4,0)$	$p = 2,\ n \equiv 7 \pmod 8$	$p = 2,\ n \equiv 3 \pmod 8$
$(3,1)$	$p = 2,\ n \equiv 0 \pmod 4$	$p = 2,\ n \equiv 0, 1 \pmod 4$
$(4,1)$	$p = 2,\ n \equiv 1 \pmod 4$	$p = 2,\ n \equiv 1 \pmod 8$
	$p = 3,\ n \equiv 6 \pmod 9$	$p = 3,\ n \equiv 6 \pmod 9$

Proof. These are the five cases of Table 1 of the common paper, with precisely the same (n, k, r, p) parameters. That result substitutes the displayed integer counts, excludes other primes by polynomial identities and checks the remaining complete residue periods. Thus the "only if" parts do not rest on a finite sample of n. □

The Johnson case of source Corollary 1 is RLCD precisely for $p = 2$ and odd n. It is CLCD precisely when, in addition, $\binom{n}{k}$ is even: the degree vanishes modulo two, so the constant-line test is decisive. This makes its printed exception exact. Domain matters: for instance $J(n, n-1) = K_n$ is RLCD when $p \mid n-1$ and CLCD when $p \nmid n-1$; the unrestricted wording of the source corollary cannot cover these endpoints by the same parity criterion. For the odd graph $KG(2k+1, k)$, the count at $j = 0$ is $P_0 = 1$ for $k \geq 2$, so it is never RLCD. For $k \geq 3$, $P_1 = 0$, which rules out CLCD by comparison with P_0. At $k = 2$ the counts are $i_0 = 0$, $i_1 = 1$, $i_2 = 3$, hence $P = (1, 1, 3)$. Thus all $P_j = \nu$ precisely in characteristic two, and $\nu = 3 \neq 0$. The binary Petersen graph is CLCD, correcting the source's universal exclusion.

For every regular graph, RLCD implies $\nu = 0$ by the diagonal of $A(A + I) = 0$, and consequently implies $C \perp \bar{C}$. Conversely, $C \perp \bar{C}$ implies RLCD exactly when $\nu = 0$. The source's example $\Gamma(13, 3, 1)$ over $\mathbb{F}_2$ has all $P_j = \nu = 1$, so is CLCD and not RLCD. This recovers the final clause and note of Corollary 1.

3.1 Every computational finding on source page 8

The following table evaluates (2) for all sixteen reported cases. The column P means the common residue of every P_j, not merely one entry; both degrees are also reduced modulo p. It is a complete small integer certificate, obtained by the displayed binomial formula, rather than an unexplained graph-code search.

n	k	r	p	P	ν	$\bar{\nu}$	RLCD	CLCD
9	4	1	2	0	0	1	yes	yes
9	4	3	2	0	0	1	yes	yes
11	4	0	2	1	1	0	no	yes
11	4	3	2	0	0	1	yes	yes
13	4	1	2	0	0	0	yes	no
13	4	3	2	0	0	0	yes	no
15	4	0	2	0	0	0	yes	no
15	4	3	2	0	0	0	yes	no
17	4	1	2	0	0	1	yes	yes
17	4	3	2	0	0	1	yes	yes
15	4	1	3	0	0	2	yes	yes
11	5	4	2	0	0	1	yes	yes
12	5	3	2	0	0	1	yes	yes
13	5	2	2	0	0	0	yes	no
13	5	4	2	0	0	0	yes	no
15	5	4	2	0	0	0	yes	no

When both answers are yes, the reflexive and complementary codes are the same orthogonal dual, as asserted in the source. All its positive findings follow, and the negative entries are now determined as well.

3.2 Spectra and the three dimension examples

We retain the classical integral spectrum of the Johnson association scheme, attributed by the source to Krebs–Shaheen, Delsarte and Brouwer–Haemers. Its multiplicities are $m_j = \binom{n}{j} - \binom{n}{j-1}$ for $0 \le j \le k$. The Kneser and Johnson eigenvalues are $(-1)^j\binom{n-k-j}{k-j}$ and $k(n-k)-j(n+1-j)$, respectively. For the general relation the retained Eberlein formula is

$$\epsilon_j = \sum_i (-1)^i \binom{j}{i}\binom{k-j}{k-r-i}\binom{n-k-j}{k-r-i}.$$

The reflexive and complement spectra follow by adding one and by subtracting from $J-I$; no further classification input is used.

For $(k,r)=(3,1)$ the eigenvalues are

$$3\binom{n-3}{2}, \quad \binom{n-4}{2} - 2(n-4), \quad -2n+11, \quad 3.$$

When $n \equiv 0 \pmod 4$, only the last two are odd. Therefore Lemma 8.3 of the common paper gives $m_2 + m_3 = \binom{n}{3} - n$, proving Example 3(1). For $(k,r)=(4,1)$ they are

$$4\binom{n-4}{3}, \quad \binom{n-5}{3} - 3\binom{n-5}{2}, \quad -(n-6)(n-9), \quad 3n-22, \quad -4.$$

When $p=2$, $n \equiv 1 \pmod 4$, only the fourth is nonzero, so the dimension is $m_3 = \binom{n}{3} - \binom{n}{2}$. When $p=3$, $n \equiv 6 \pmod 9$, exactly the fourth and fifth survive, giving $m_3 + m_4 = \binom{n}{4} - \binom{n}{2}$. The degree-$4\binom{n-4}{3}$ row is a nonzero word and supplies the two stated distance upper bounds. These are source Example 3(2)–(3), with no claim that either upper bound is always attained.

4 Incidence spaces, parameters and retained distances

For an s-subset $\Lambda \subseteq \Omega$, let w_Λ be the indicator of the k-subsets containing Λ; put $W_s = \langle w_\Lambda : |\Lambda| = s \rangle$ and $E(W_s) = \langle w_\Lambda - w_{\Lambda'} \rangle$. These are the spaces of source equation (4). The alternating transversal word associated with k disjoint pairs, from source equation (5), belongs to $W_s^\perp$ for $s < k$: for any fixed s-set, either no transversal contains it, or at least one pair is unspecified and its two choices contribute opposite signs. Thus $W_\Pi \subseteq W_s^\perp$ over every characteristic. This elementary cancellation is all that is needed of that notation here.

Proposition 2. *Suppose $1 \le k < n$. Then $\dim W_1 = n$ if $p \nmid k$, and $n-1$ if $p \mid k$. In the binary case set*

$$b_r = \sum_{i \text{ odd}} \binom{r}{i}\binom{n-r}{k-i}, \qquad \epsilon = [k \text{ is even}].$$

The full Hamming weight enumerator, including repeated weights and the zero word, is

$$\sum_{c \in W_1} z^{\mathrm{wt}(c)} = 2^{-\epsilon} \sum_{r=0}^{n} \binom{n}{r} z^{b_r}. \tag{3}$$

Proof. Map $(a_x)_{x\in\Omega}$ to $\sum_x a_x w_x$. If this is zero, compare its coordinates on $T \cup \{x\}$ and $T \cup \{y\}$, where $|T| = k-1$ and $x, y \notin T$. Such a T exists for every $x \ne y$ because $k < n$. It follows that $a_x = a_y$. Hence the kernel consists exactly of constant coefficient vectors $a\mathbf{j}$ with $ka = 0$, proving the dimension. In characteristic two a coefficient support S of size r gives a nonzero coordinate at X precisely when $|S \cap X|$ is odd, and the count is b_r. Each word has exactly 2^ϵ coefficient preimages. Dividing the sum over all coefficient supports by that constant fiber size proves (3). □

This recovers and qualifies source Result 4. The dimension statement requires $k < n$, as in all its graph applications. Its displayed weight values follow from $b_{n-r} = b_r$ for even k and $b_{n-r} = \binom{n}{k} - b_r$ for odd k. Its unqualified claim of $\binom{n}{r}$ words of weight b_r must instead respect fibers and collisions. For example, when $(n,k) = (8,2)$, weight 16 has $\binom{8}{4}/2 = 35$ words, not 70. Formula (3) also includes the constant word when k is odd. These corrections refer to the statement printed in F017; they are not assertions about its cited predecessor.

4.1 The complete quoted parameter list

Source Result 3 quotes earlier code classifications, principally Key–Moori–Rodrigues and Fish–Key–Mwambene, listed as source references [11,14,16,17,21–23]. The following table preserves every listed case. For $k = 3$ take $n \ge 7$ and $N = \binom{n}{3}$; it gives the reflexive space, its dimension and distance, and those of its adjacency dual.

condition	$RC_r = C_r^\perp$	$\dim RC_r$	$d(RC_r)$	$d(C_r)$
$r=0, p \ge 5, n \equiv 5 \pmod p$	W_2	$\binom{n}{2}$	$n-2$	≤ 8
$r=0, p=2, n \equiv 1 \pmod 4$	$W_1 + W_2$	$\binom{n}{2}$	$n-2$	8
$r=0, p=3, n \equiv 5 \pmod 9$	$W_2 + \langle \mathbf{j} \rangle$	$\binom{n}{2}$	$n-2$	8
$r=1, p=2, n \equiv 0 \pmod 4$	W_1	n	$\binom{n-1}{2}$	4
$r=2, p=2, n$ odd	W_2	$\binom{n-1}{2}$	$n-2$	4

In every row $\dim C_r = N - \dim RC_r$. The reflexive duality in all five rows follows from the common scalar classification; it is not a retained computational assertion. The final row, including both distances and both dimensions, is proved by Theorem 3.2 of the common paper and Theorem 3.3 of the common paper. In the third row Theorem 4.2 of the common paper proves the space identity and dimension: $\dim W_2 = \binom{n}{2} - 1$ and $\mathbf{j} \notin W_2$. Thus the two printed

dimensions $\binom{n}{2}+1$ and $N-\binom{n}{2}-1$ need the indicated one-unit correction. The final row similarly corrects the printed $\binom{n}{2}$ to $\binom{n-1}{2}$. The remaining space identifications and distances in the first four rows are retained, exactly as quoted by the source from its earlier papers; the common projector proof alone is not a distance proof. This distinction includes the distance 8 in the ternary row. The first row's upper bound 8 also follows directly from a nonzero alternating transversal word on six points, using $W_{\Pi} \subseteq W_2^{\perp}$.

For $k=2, n \geq 5, p=2$, source Result 3 has two additional cases. If $n \equiv 3 \pmod 4$, the identification $RC_0 = W_1 + \langle \mathbf{j} \rangle$ has parameters $[\binom{n}{2}, n, n-1]_2$, with dual $[\binom{n}{2}, \binom{n}{2}-n, 4]_2$. Indeed $RC_0 = \mathrm{im}(J + A_{J(n,2)})$; the two summands are annihilating projectors onto $\langle \mathbf{j} \rangle$ and W_1, respectively, since $\binom{n}{2}$ is odd and the Johnson degree is even. Their images therefore add directly. The two distances just stated remain the source's quoted earlier-code results. For either odd residue of n, the Johnson code is $C_1 = W_1$, with parameters $[\binom{n}{2}, n-1, n-1]_2$; its reflexive dual has parameters $[\binom{n}{2}, \binom{n}{2}-n+1, 3]_2$. The missing $+1$ in the printed dual dimension is corrected by the simplex ranks. These Johnson parameters and distances are fully proved in the common paper. If $n \equiv 1 \pmod 4$, the quoted $C_0 = E(W_1)^{\perp}$ is LCD. Here $E(W_1) = W_1$: the coefficient kernel is the odd-weight constant vector, so every coefficient class has an even-weight representative. The common complement theorem, applied to the complementary Johnson graph whose code is W_1, gives $C_0 = W_1^{\perp}$ and proves LCD.

Source Result 5 is a separately quoted incidence theorem: for $n \geq 2k+1$, W_{k-1} has minimum weight $n-k+1$ over every prime field; for $n > 2k+1$ its minimum words are scalar multiples of the stars w_{Λ}; its dimension is at least $\binom{n-1}{k-1}$. The binary equality, minimum weight and minimum-word statement are proved by Lemma 3.1 of the common paper and Theorem 3.3 of the common paper. The odd-characteristic clauses remain retained source reference [11] inputs. Likewise the minimum $\binom{n-1}{2}$ for the binary W_1 on triples is retained from the quoted Result 4; the exact enumerator above gives an alternative reduction to an integer inequality but is not silently substituted for that inequality.

4.2 All four Johnson parity cases and decoding

Put $E = u_{k-1}d_k$ on binary k-subsets, in the notation of the common simplex section. The identity $A = E + (k \bmod 2)I$, with $E^2 = nE$, proves all four cases of source Result 6. For odd n, E is the projector onto W_{k-1}: $C = W_{k-1}^{\perp}, RC = W_{k-1}$ for odd k, and the two are interchanged for even k. Both hulls vanish. For even n, $\mathrm{im}\, E \subseteq \mathrm{Hull}(W_{k-1})$; the adjacency and reflexive codes are $(\mathrm{im}\, E, V_k)$ for even k and $(V_k, \mathrm{im}\, E)$ for odd k. These are consequences of Theorem 3.2 of the common paper, without the source's earlier-code citation as an extra premise.

In source Example 1, $n > 7$, $n \equiv 0 \pmod 4$, let $C = RC_1 = W_1$. For a received binary vector v with support S,

$$v = \sum_{x \in S} s_x^1 + \sum_{x \in S} r_x^1 = v_C + v_{\perp}.$$

The preceding quoted parameters give minimum distance $\binom{n-1}{2}$. A coset leader c of $C - v_{\perp}$ yields the decoded word $v + c$, uniquely if fewer than $\frac{1}{2}\binom{n-1}{2}$ errors occurred. This is exactly the nearest-word argument of the projection section. The source's suggestion that a faster explicit decoder would be desirable is recorded as a question; no new decoding-complexity claim is inferred from this decomposition.

5 Strongly regular graphs

Let Γ be noncomplete and nonempty, strongly regular with parameters (N, k, λ, μ). The three kinds of adjacency entries give

$$A^2 = (k-\mu)I + (\lambda-\mu)A + \mu J.$$

Thus Proposition 1.5 of the common paper recovers both parts of source Proposition 4:

$$\begin{aligned} C \text{ is RLCD} &\iff k = 0, \quad \lambda = -1, \quad \mu = 0 \pmod p, \\ C \perp \bar{C} &\iff k-\mu = 0, \quad k-1-\lambda = 0 \pmod p. \end{aligned}$$

In the second line CLCD holds exactly when k and $N-1-k$ do not both vanish. This completes the possible-exception clause in the source. In the first line the nonconstant eigenvalues, when integral, have residues zero and minus one: their product is $\mu - k = 0$ and their sum is $\lambda - \mu = -1$. Hence the dimension is the multiplicity of the minus-one residue. The radical in the usual formula is

$$\Delta = (\lambda-\mu)^2 + 4(k-\mu),$$

with a plus sign. The minus sign printed in Proposition 4(1) is a transcription error, inconsistent with the preceding source spectrum. The classical multiplicities used here are

$$m_\pm = \frac{1}{2}\left(N - 1 \pm \frac{(N-1)(\mu-\lambda) - 2k}{\sqrt{\Delta}}\right), \qquad \rho_\pm = \frac{\lambda - \mu \pm \sqrt{\Delta}}{2}.$$

They follow from $m_+ + m_- = N-1$ and $k + m_+\rho_+ + m_-\rho_- = 0$. The source's cited conference-parameter alternative has $\lambda = \mu - 1$, $k = 2\mu$, $N = 4\mu + 1$ and N a sum of two integer squares; the other spectral alternative has integral eigenvalues. That classical feasibility statement is retained from the source's Cameron–van Lint reference, not reproved here or needed to test the displayed graph families. On each nonconstant eigenspace the complement eigenvalue is $-1-\rho$, with the same multiplicity. In particular $T(m)$ has nonconstant eigenvalues $m-4, -2$, of multiplicities $m-1, m(m-3)/2$, and its complement has $3-m, 1$ on those respective spaces. This corrects the mismatched complement entries in the source's triangular spectral display.

Proposition 3. *The four instances in source Corollary 2 have the following conclusions.*

1. *For odd $m \geq 5$, the binary triangular graph $T(m) = J(m, 2)$ is RLCD, of dimension $m-1$. It is CLCD exactly when $m \equiv 1 \pmod 4$. Its complement $KG(m, 2)$ is RLCD exactly for $p = 2$ and $m \equiv 3 \pmod 4$.*

2. *For odd $m \geq 3$, the binary lattice graph $L_2(m)$ and its complement are both RLCD, of dimensions $2m-2$ and $(m-1)^2$.*

3. *Let $q \equiv 1 \pmod 4$ be any odd prime power. The Paley graph $P(q)$ is RLCD over $\mathbb{F}_p$ exactly when $p = 2$ and $q \equiv 1 \pmod 8$, or when p is odd and $q \equiv 1 \pmod p$. In each case the dimension is $(q-1)/2$, including when q is not a square.*

4. *Every strongly regular graph with parameters $(81, 20, 1, 6)$ is binary RLCD of dimension 20. The specified generalized Paley graph below realizes the source example.*

Proof. For $T(m)$, the parameters are $\binom{m}{2}, 2(m-2), m-2, 4)$. The reflexive test forces $p = 2$ and odd m. The simplex theorem gives the dimension. In that characteristic its degree vanishes and the exact defect theorem gives CLCD precisely when $\binom{m}{2}$ is even, namely $m \equiv 1 \pmod 4$. Substituting the complement parameters gives the final condition; this also follows from the complete subset classification in the common paper.

For $L_2(m)$, the parameters are $(m^2, 2(m-1), m-2, 2)$, and the test gives $p = 2$ and odd m. Its nonconstant eigenvalues are $m-2$ and -2, of multiplicities $2(m-1)$ and $(m-1)^2$. The LCD rank lemma gives the first dimension. Equation (1) proves that the complement is also RLCD. Both degrees vanish modulo two, so the exact defect gives its dimension $m^2 - 1 - (2m-2) = (m-1)^2$.

Paley parameters are $(q, (q-1)/2, (q-5)/4, (q-1)/4)$, so the displayed conditions are exactly the reflexive test. The graph is self-complementary: multiplication of $\mathbb{F}_q$ by any nonsquare interchanges square and nonsquare differences and hence the two adjacency matrices. Their ranks are equal over every field. In each stated characteristic both degrees vanish, so the constant-line defect gives $2 \operatorname{rank} A = q - 1$. This proves the dimension without integral eigenvalues.

For $(81, 20, 1, 6)$, the binary reflexive test holds. The two nonconstant eigenvalues are 2 and -7, of multiplicities 60 and 20. Only the latter is odd, and the rank lemma proves the final assertion. □

The displayed triangular exceptional case in the source is genuine: $m \equiv 3 \pmod 4$ gives both degrees zero and a one-dimensional missing constant line. The needed congruence is $\binom{m}{2} \equiv 1 \pmod 2$; the stronger modulus-four assertion printed there does not hold for every such m. The binary Petersen graph is not RLCD, but, as explained by the subset classification, it is CLCD.

For the symplectic graph on projective points of a $2m$-dimensional symplectic space, $m \geq 2$, put

$$N = \frac{q^{2m}-1}{q-1}, \quad k = \frac{q^{2m-1}-1}{q-1} - 1, \quad \mu = \frac{q^{2m-2}-1}{q-1}, \quad \lambda = \mu - 2.$$

No field characteristic can make this graph RLCD: the requirements $\mu = 0$ and $\lambda = -1$ would give $-2 = -1$. Nor can it be CLCD, since $k - 1 - \lambda = (k - \mu) + 1$ makes the two complement tests incompatible. For its complement the adjacent and nonadjacent intersection numbers are equal, so its reflexive tests are also incompatible. These direct parameter arguments establish all exclusions in the source. For completeness the nonconstant eigenvalues are $q^{m-1} - 1$ and $-q^{m-1} - 1$, with multiplicities

$$\tfrac{1}{2}(N - 1 + q^m), \qquad \tfrac{1}{2}(N - 1 - q^m).$$

The source prints $\mu - 1$ instead of $N - 1$ in both multiplicities; the correction is forced by their sum $N - 1$ and the trace equation. No erroneous spectral multiplicity is needed for the exclusion proof.

6 The complete generalized Paley table

To avoid confusing the subgroup index with the valency, write $\mathrm{GP}(q, h)$ for the graph on $\mathbb{F}_q$ with connection subgroup $S = \langle g^h \rangle$, where g generates $\mathbb{F}_q^\times$, $h \mid q-1$ and $-1 \in S$. Its degree is $\nu = (q-1)/h$. Translations preserve the graph and both adjacency and reflexive codes. Thus the exact group hypothesis of Theorem 8.1 of the common paper is available for every row below.

Theorem 4. *All twelve rows of source Table 1 have the following parameters. Every displayed adjacency code is RLCD and none is CLCD. A star in the last column means that its complete set of minimum words is exactly the set of adjacency rows.*

q	h	p	(ν,λ,μ)	C_A	C_{I+A}	
25	*3*	*2*	$(8,3,2)$	$[25,8,8]$	$[25,17,4]$	$*$
49	*4*	*2*	$(12,5,2)$	$[49,12,12]$	$[49,37,4]$	$*$
81	*4*	*2*	$(20,1,6)$	$[81,20,20]$	$[81,61,6]$	$*$
81	*5*	*2*	$(16,7,2)$	$[81,16,16]$	$[81,65,4]$	$*$
121	*3*	*2*	$(40,15,12)$	$[121,40,20]$	$[121,81,8]$	
121	*4*	*2*	$(30,11,6)$	$[121,90,6]$	$[121,31,11]$	
121	*4*	*3*	$(30,11,6)$	$[121,30,20]$	$[121,91,6]$	
121	*6*	*2*	$(20,9,2)$	$[121,20,20]$	$[121,101,4]$	$*$
243	*11*	*2*	$(22,1,2)$	$[243,110,d]$	$[243,133,\delta]$	
729	*4*	*2*	$(182,55,42)$	$[729,546,d]$	$[729,183,\delta]$	
729	*7*	*2*	$(104,31,12)$	$[729,104,d]$	$[729,625,\delta]$	
729	*14*	*2*	$(52,25,2)$	$[729,52,d]$	$[729,677,\delta]$	

In the last four rows only $d \le \nu$ and $\delta \le \nu+1$ are asserted. Every bracket describes a code over $\mathbb{F}_p$. The printed degree 201 *in the* $(81,4)$ *source row is corrected to* 20*.*

Proof. The complete finite certificate constructs each field as $\mathbb{F}_\ell[X]/(f)$, recording f, the primitive element and the subgroup S. It verifies irreducibility of f and all $q-1$ distinct nonzero powers of the generator. For every nonzero t, it counts $|S \cap (t+S)|$. Translation reduces every pair of distinct vertices to $(0,t)$, so those counts prove the displayed strongly regular parameters, not just a random selection of them.

In each stated characteristic $\nu = \mu = 0$ and $\lambda = -1$. The common SRG criterion gives $A^2 = -A$, proving reflexive duality. The complement is orthogonal, but both degrees vanish, since $q-1 = 0$ modulo p in every row. The constant-line defect therefore proves that none is CLCD. The two nonconstant rational eigenvalues are the roots of $X^2 - (\lambda-\mu)X - (\nu-\mu)$; their multiplicities satisfy $f+g = q-1$ and $\nu + fr + gs = 0$. They are integral in every row. Lemma 8.3 of the common paper now gives all dimensions, with Gaussian elimination also providing systematic generators for the first eight rows.

For each of the first eight rows the certificate supplies, for both codes, a word of the displayed distance D and a complete enumeration of nonzero messages of weight at most $\lfloor K(D-1)/q \rfloor$, where K is that code's dimension. Every enumerated word has weight at least D. The translation action just established allows Theorem 8.1 of the common paper to prove all sixteen exact distances. For the five starred rows a second complete enumeration, with bound $\lfloor KD/q \rfloor$, finds only adjacency rows at weight D. Their set is translation invariant, so Corollary 8.2 of the common paper proves that these are all minimum words. For each of the last four rows, an adjacency row has weight ν and a reflexive row has weight $\nu+1$, proving exactly the asserted upper bounds and no stronger conclusion. □

The finite premise consists of the whole programs `check_paley_codes.py` and `enumerate_sparse_messages.c`, together with their complete field, basis, witness and enumeration records. The Python driver checks systematic bases, membership of each witness, the exact binomial message count and the lower-bound output. The C enumerator visits every strictly increasing tuple of row positions and every tuple of nonzero coefficients once. Its binary and ternary arithmetic and boundary conditions have a separate accepted review against coordinate oracles. Those checks are required evidence for the finite distance conclusions; they are not replaced by a reference to the historical Magma output. The four large rows have no unreported minimum-distance computation.

The source attributes the original generalized Paley computations to J. and P. Limbupasiriporn (its reference [27]), and cites Seneviratne–Limbupasiriporn [30] for the neighbour formula. Indeed multiplication by an element of S makes $|N(1) \cap S|$ the common adjacent-pair count for every such graph. The source's broader assertion from [27] that $\lambda - \mu$ is odd for strongly regular cases with $h > 2$ is a retained cited observation; our certificate verifies it in

all twelve displayed rows, and neither the generic SRG argument nor this table purports to classify all generalized Paley graphs.

7 Other decompositions and the concluding examples

7.1 Binary triples and ternary triples

For binary $J(n,3)$ with odd n, the simplex identity gives, for $x=\{a,b,c\}$,

$$e_x = r_x^2 + w_{\{a,b\}} + w_{\{a,c\}} + w_{\{b,c\}}.$$

The first term is in $C = W_2^{\perp}$ and the remaining term is in W_2. This is the first decomposition of source Section 6; summing against the coordinates of any vector gives its orthogonal splitting. The source also records a typographical correction in its predecessor [21]; that is historical attribution, not a newly checked claim about that predecessor's printed pages.

For ternary triples and $n \equiv 1 \pmod 3$, the second displayed identity is

$$e_{\{a,b,c\}} = \mathbf{j} - r_x^0 + r_x^1 + w_a + w_b + w_c. \tag{4}$$

At a triple meeting x in $t=0,1,2,3$ points, the right side has coefficient $1-[t=0]+[t=1]+t$, namely $0,0,0,1$ modulo three. This proves the literal identity directly. The membership $w_a \in C_0^{\perp}$ also follows by counting: for a triple disjoint from a, the relevant count is $\binom{n-4}{2} = 0$ modulo three, and for a triple containing a it is zero. The additional assertions $r_x^1 \in C_0$ and $\mathbf{j} \in C_0$ or $C_0^{\perp}$ are retained from the source's explicit invocation of Key–Moori–Rodrigues [23], Proposition 18. With those memberships, summing (4) gives exactly the source's splitting of every vector. In particular $C_0 + C_0^{\perp}$ is the whole space, so the dimension argument proves LCD and makes that splitting orthogonal and unique. This application does not present that borrowed membership theorem as a consequence of the unrelated $n \equiv 5 \pmod 9$ case.

7.2 A uniform odd-graph correction

Proposition 5. *For every $k \geq 1$, let A be the binary adjacency matrix of $KG(2k+1,k)$ and put $E = u_{k-1}d_k$. Then*

$$A^2 = I + E, \qquad A^3 = A, \qquad C_A = W_{k-1}^{\perp}, \qquad C_A^{\perp} = W_{k-1}.$$

In particular C_A is LCD. Its orthogonal decomposition on a unit vector is

$$e_x = \sum_{y \cap x = \varnothing,\ |y|=k} r_y + \sum_{R \subset x,\ |R|=k-1} w_R. \tag{5}$$

Moreover $\mathbf{j} \in C_A$ for even k and $\mathbf{j} \in C_A^{\perp}$ for odd k.

Proof. For distinct k-sets with intersection size j, the common-neighbour count is $\binom{1+j}{k}$: it is one precisely at $j = k-1$. The diagonal count is $k+1$. Thus $A^2 = J(2k+1,k)_{\mathrm{adj}} + (k+1)I = I + E$ by the common Johnson identity. For a fixed $(k-1)$-set R, the T-coordinate of Aw_R counts k-sets containing R and disjoint from T. If $R \cap T \neq \varnothing$ it is zero; otherwise there are exactly two choices for the remaining point. Therefore $Au_{k-1} = 0$, and $AE = 0$. Multiplying the first identity by A proves $A^3 = A$.

It follows that $\operatorname{im} A = \operatorname{im} A^2$: one inclusion is automatic and $Av = A^2(Av)$ proves the other. Since $2k+1$ is odd, Theorem 3.2 of the common paper makes E the orthogonal projector onto W_{k-1} and $I+E$ the projector onto its dual. This proves the code identities and LCD. Applying $A^2 + E = I$ to e_x gives (5); its two summands lie in the asserted orthogonal spaces. Finally $d_k\mathbf{j} = (k+2)\mathbf{j}$ and $u_{k-1}\mathbf{j} = k\mathbf{j}$, so $E\mathbf{j} = k\mathbf{j}$ in characteristic two. This proves the constant-word clause. □

This also gives dimension $\binom{2k}{k}$ and minimum distance $k+1$, with all minimum words the $(k+1)$-set boundaries, by Theorem 3.3 of the common paper. The source's needed conclusion is LCD for all k, which now has a uniform proof. Its formula on physical page 15 is not literally correct: at $k=2$ the first printed sum is empty and $k\mathbf{j}=0$, leaving only A^2e_x. At a different pair meeting x in one point this has coefficient one, whereas e_x has coefficient zero. Equation (5) supplies the missing complementary term. This refutes that formula as printed in F017, not the LCD conclusion and not an unexamined assertion in the cited Fish–Fray–Mwambene paper.

7.3 All Hamming examples, with their scope

Write $H_k(d,q)$ for the graph on q-ary words of length d, joined when their Hamming distance is k. The common distance-count formula Equation (5.2) of the common paper and scalar test cover the concluding examples. For odd q, the one-coordinate off-diagonal operators are commuting idempotents over $\mathbb{F}_2$, and each distance-k operator is the corresponding elementary symmetric polynomial. Hence every $H_k(d,q)$ is binary RLCD by Corollary 5.2 of the common paper. In particular this proves both reported $H_1(6,3)$ and $H_2(6,3)$ cases.

For $H_2(d,2)$ in characteristic three the only nonzero common-neighbour counts can occur at distances $0,2,4$, and are $\binom{d}{2}$, $2(d-2)$ and 6 respectively. Adding the adjacency term at distance two makes all entries zero when $d\equiv 0 \pmod 3$. Thus this entire family is RLCD, including $d=6,9$ from the source. No unverified classification of all primes, alphabets or distances is inferred from these examples.

For the binary cube $H_1(d,2)$ with positive even d, the common cube theorem gives $A^2=0$, rank 2^{d-1}, and $C_A=C_A^{\perp}$. Its complement matrix is invertible: $B^2=I$ in characteristic two. Consequently the printed complementary duality assertion for $d=6,8,10$ must be corrected to self-duality of the adjacency code. It is not an RLCD or CLCD example.

The concluding references to McLaughlin subgraphs and Hamming papers are source context. The former is the separate L050 application, with its own full finite proof obligations. The latter includes an "in preparation" item in the 2017 bibliography; we do not turn that historical status into a present-day open-problem claim. The source's prospective search for more RLCD and CLCD families and for efficient decoders remains outside the proved contract except for the uniform families explicitly established above.

8 Attribution, finite premises and formal scope

All graph and code results are attributed to the source and the predecessors it names. The common paper gives complete proofs of the projector, scalar-count, binary incidence, ternary dimension and transitive-enumeration arguments. Its complete subset-residue certificate and the two complete Paley programs and their records remain actual finite premises. Optional small checks of the local identities and counterexamples are inventoried separately. None of these programs is silently omitted from the proof boundary because a written-text comparison excludes software symmetrically.

The common six generic projection declarations and three generic transitive-distance deductions are the existing scoped Lean evidence. They do not formalize the graph families, the source-specific incidence bridges, retained spectra and distances, or the finite computations. No further formalization is claimed by this application. The exact source-contract and dependency records accompany the paper; unknown external proof, historical software and graphic costs are not treated as zero.

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